Module 2: Banco Federal de Finanças

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I. Background

Banco Federal de Financas recently concluded an ineffective telemarketing campaign. The bank hired us to demonstrate how machine learning can improve customer contact in future campaigns.

This case study has three primary objectives:

- **Factor Identification:** Identifying the factors most associated with a customer subscribing
- **Predictive Targeting:** Building a model that identifies the customers most likely to respond to future campaigns
- **Value Demonstration:** Showing that model-based targeting reaches subscribers at a much higher rate than randomly selected customers

Meeting these goals involves analyzing data from the previous campaign, building a model to find patterns in it, and using those patterns to predict which groups to target in future outreach.

II. Methodology

We used a supervised model to identify the conditions associated with whether a customer subscribes, represented by the feature "y." This approach fits our data well, since the training set includes several thousand observed target values.

The dataset contains three categories of features, each serving a different purpose:

Demographic data shows which groups of people have subscribed in the past.
Contact and timing details indicate the best way and moment to reach customers. These include economic indicators, which show whether customers are more likely to subscribe under certain economic conditions.
Previous campaign data shows how earlier interactions relate to later responses.

The model has two main limitations. Because the data is historical, shifts in customer preferences and the economy may reduce its effectiveness over time. The economic features are also closely correlated with one another. They help explain differences in success between past campaigns, but they offer less guidance for future action.

Before building the decision tree, we first checked which features were most important. **Figure 1** shows the results.

The most significant features are the quarterly number of employees at the bank (‘nr.employed’) and the 3-month Euribor rate (‘euribor3m’). Both reflect the economy at the time of the previous customer contact. In short, the state of the economy is the strongest

predictor of whether a customer will agree to the current campaign. This is intuitive: customers who are already struggling financially are unlikely to take on perceived financial risk.

Surprisingly, the month of contact is one of the most important features tied directly to customer compliance, ranking fourth in the figure. General trends in mood, such as seasonal depression, could explain this, but the model does not say why the feature matters. Further research is needed to understand why the month affects compliance so heavily.

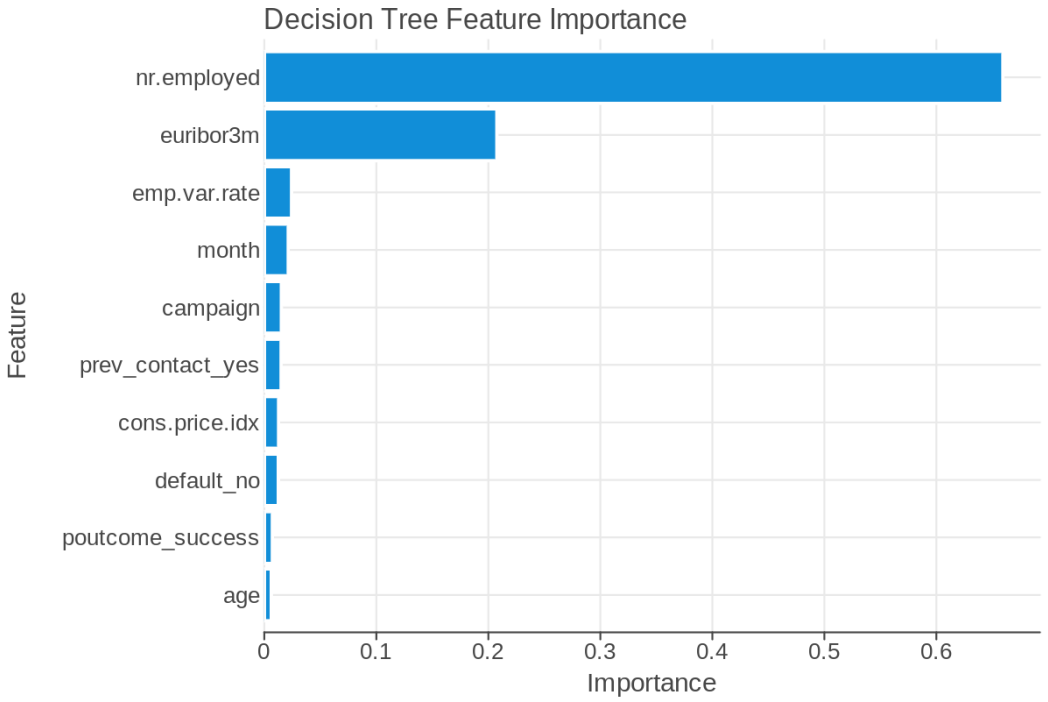


Figure 1: Decision Tree Feature Importance

To train the model, we split the dataset of around 37,000 samples into 70% training and 30% testing data, which leaves roughly 11,100 samples for testing. This split balances two needs: enough data to train the model and enough held out to evaluate its performance reliably.

III. Model Results

Figure 2 shows the confusion matrix for the test set, with "yes" (the customer subscribed) as the positive class. Of the 1,262 subscribers, the model identified 848 and missed 414. Of the 2,628 customers it predicted would subscribe, 848 did and 1,780 did not. **Table 1** summarizes the scores.

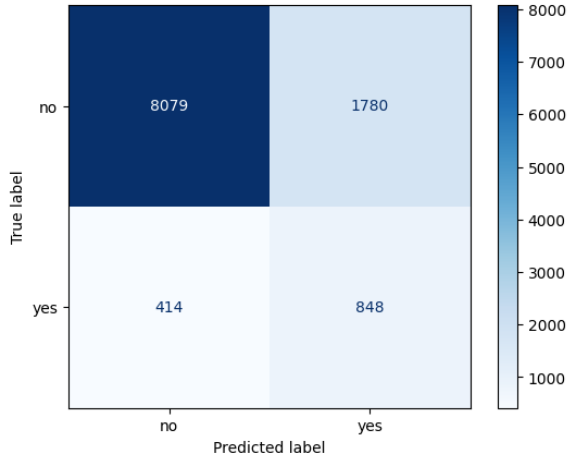


Figure 2: Confusion Matrix for the Test Set

Metric	Formula	Score
Accuracy	(TP + TN) / Total	80.3%
Precision	TP / (TP + FP)	32.3%
Recall	TP / (TP + FN)	67.2%
F1 score	2PR / (P + R)	43.6%

Table 1: Classification Metrics for the Test Set

Accuracy is 80.3%, below the 88.7% that a model always predicting "no" would reach. This is expected, because the model uses balanced class weights and favors finding subscribers over avoiding false alarms. Recall of 67.2% means it finds about two of every three subscribers. Precision of 32.3% means about one in three predicted subscribers actually subscribes, which is still roughly 2.8 times the random rate of 11.4%, or about one call in nine. The F1 score is 43.6%. The trade-off is 1,780 calls to customers who did not subscribe. Adjusting the classification threshold or tree depth could shift this balance.

IV. Python Notebooks

The notebook used for this case study is available as a GitHub Gist:

Google Colab

V. Discussion Responses

The data used to train the model complies with GDPR. It contains no personal information and cannot be used to track anyone. The information is vague enough that neither the model nor an external user could identify an individual from it alone. For example, the data includes no names, ethnicity, birth location or date, place of residence, government identifiers, contact details, or account specifics.

We also examined whether prior outreach affects a customer's decision to join the current campaign, mainly out of concern that contacting customers too often might backfire. **Figure 3** shows that previous contact has no large or meaningful effect either way. The share of customers who agreed is roughly the same whether or not they were previously contacted.

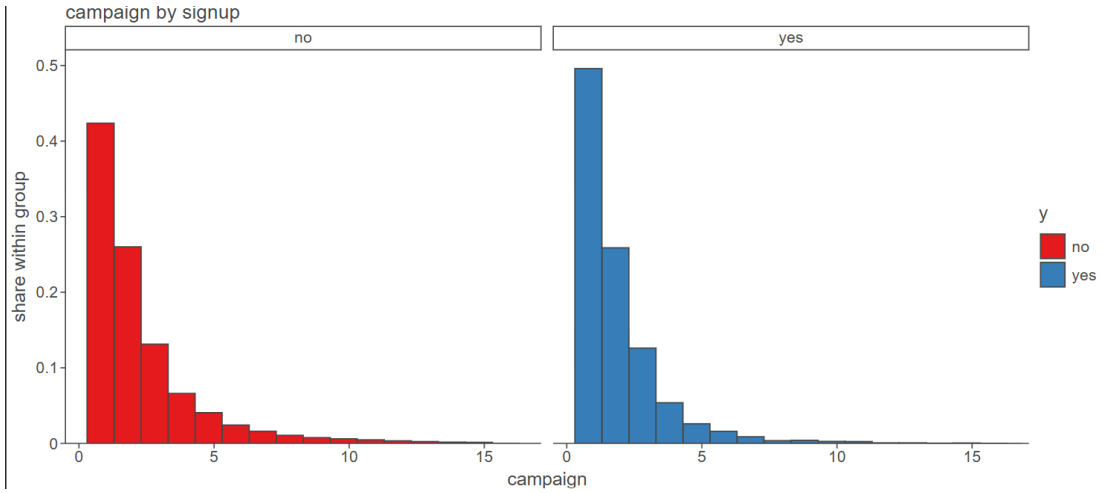


Figure 3: Comparison of Previous Contact to Agreement of Campaign

A machine learning model is better than random calling because it finds patterns in thousands of customer records that a person could not review by hand. On our test set, only about 11.4% of customers subscribed, so random calling would reach a subscriber roughly one call in nine. When our model predicted that a customer would subscribe, it was right 32.3% of the time, about 2.8 times the random rate, and it found 67.2% of all subscribers. The trade-off is that it also flagged 1,780 customers who did not subscribe, so some calls will still be wasted. Even so, calling the customers the model selects would give the bank more subscribers per call than random outreach. Because the model relies mainly on economic indicators, it should be retrained as conditions change before it is used for future campaigns.